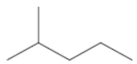
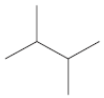
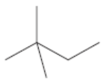
<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td>7.</td><td>(a)</td><td>(i)</td><td></td><td>compounds with the same molecular formula but a different structure / structural formula</td><td>1</td><td></td><td></td><td>1</td><td></td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td></td><td>
<table>
<tr><th>Name</th><th>Shortened structural formula</th><th>Skeletal formula</th><th>Boiling temperature / °C</th></tr>
<tr><td>hexane</td><td>$CH_3CH_2CH_2CH_2CH_2CH_3$</td><td></td><td>69</td></tr>
<tr><td>2-methylpentane</td><td>$CH_3CH(CH_3)CH_2CH_2CH_3$</td><td></td><td>62</td></tr>
<tr><td>3-methylpentane</td><td>$CH_3CH_2CH(CH_3)CH_2CH_3$</td><td></td><td>63</td></tr>
<tr><td>2,3-dimethylbutane</td><td>$(CH_3)_2CHCH(CH_3)_2$</td><td></td><td>58</td></tr>
<tr><td>2,2-dimethylbutane</td><td>$CH_3C(CH_3)_2CH_2CH_3$</td><td></td><td>50</td></tr>
</table>
award (1) for each correct answer
accept $(CH_3)_2CHCH_2CH_2CH_3$ for 2-methylpentane
</td><td></td><td>3</td><td></td><td>3</td><td></td><td></td></tr>
</table>

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		as the main carbon chain length increases, the boiling temperature increases (or converse) (1) award (1) for either of following lengthening the chain increases the Van der Waals forces between molecules more heat energy is required to overcome the increased forces (or converse)		1	1	2		
	(b)	(i)		C_6H_{14} + $9\frac{1}{2}O_2$ $\rightarrow$ $6CO_2$ + $7H_2O$ accept $2C_6H_{14}$ + $19O_2$ $\rightarrow$ $12CO_2$ + $14H_2O$ formulae (1) balancing (1) – only awarded if all formulae are correct		2		2		
		(ii)		the number and types of bond are the same (1) the energy absorbed by breaking bonds and the energy released by forming bonds will be similar (1)		2		2		
		(iii)		it has the lowest boiling point / is the most volatile / vaporises most quickly accept explanation of lower boiling point		1		1		
		(iv)	I	energy absorbed = (5 × 348) + (14 × 413) + (6.5 × 495) = 10739.5 (1) energy released = (6 × 1072) + (14 × 464) = 12928 (1) enthalpy change = 10739.5 – 12928 = –2188.5 (1)		3		3	2	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
			II	award (2) for quantitative comparison e.g. about twice as much energy is released (with excess oxygen) / additional 1971.5 kJ mol^{-1} released (with excess oxygen) award (1) for simple statement e.g. the fuel releases more energy (with excess oxygen) ecf possible from part **I**			2	2		
			III	award (1) for any reference to CO/carbon monoxide	1			1		
				Question 7 total	**2**	**12**	**3**	**17**	**2**	**0**